

VLAN Concepts Theory: VLANs: Segmentation, benefits, and use cases. Practical : Create VLANs on a switch and assign ports. Task: Document VLAN configurations.

Task 1.

On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with Ctrl-Z to exit.
Switch(config)#
Switch(config)#vlan 10 engg
Switch(config)#^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#name engineering
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name marketing
Switch(config-vlan)#exit
Switch(config)#do wr
Building configuration...
```

Task 2.

Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

```
Switch(config)#int range Fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#int range Fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

Task 3.

Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

1. VLAN Details

enable

conf t

vlan 10

name engineering

ex

do wr

vlan 20

name marketing

ex

do wr

2. Cisco CLI Configuration

enable

configure terminal

vlan 10

name Engineering

exit

vlan 20

name Marketing

exit

do wr

```
interface range fastEthernet 0/1-2
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

```
do wr
```

```
interface range fastEthernet 0/3-4
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

```
do wr
```

Task 4.

Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

1. Network separation:

The coding lab and networking lab can be placed in different VLANs, keeping their broadcast traffic separated and reducing unnecessary traffic.

2. Better security and management:

VLANs make it easier to control which devices or labs can communicate with each other. For example, access between the coding and networking labs can be controlled through routing and access-control rules.